# TABBIE AI

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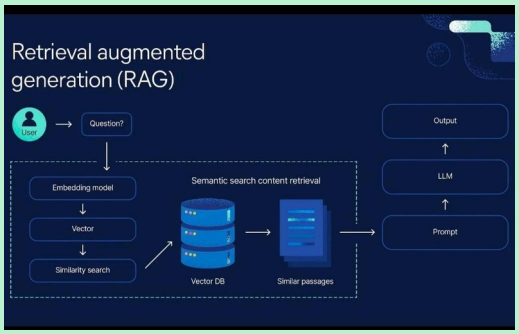


## Introduction

In modern data-driven environments, accessing database insights typically requires knowledge of Structured Query Language (SQL). This creates a significant barrier for non-technical users who need to extract information from databases quickly and .efficiently

To address this challenge, Tabbie AI introduces an intelligent Natural Language to SQL (NL2SQL) middleware that allows users to interact with databases using simple natural language queries. Instead of writing complex SQL commands, users

can ask questions in everyday language and receive .accurate results instantly

The system leverages advanced AI technologies, including Large Language Models (LLMs) and vector-based semantic search, to understand user intent and translate it into executable SQL queries. By combining these technologies with a hybrid memory architecture, Tabbie AI is able to deliver both high accuracy and high perform

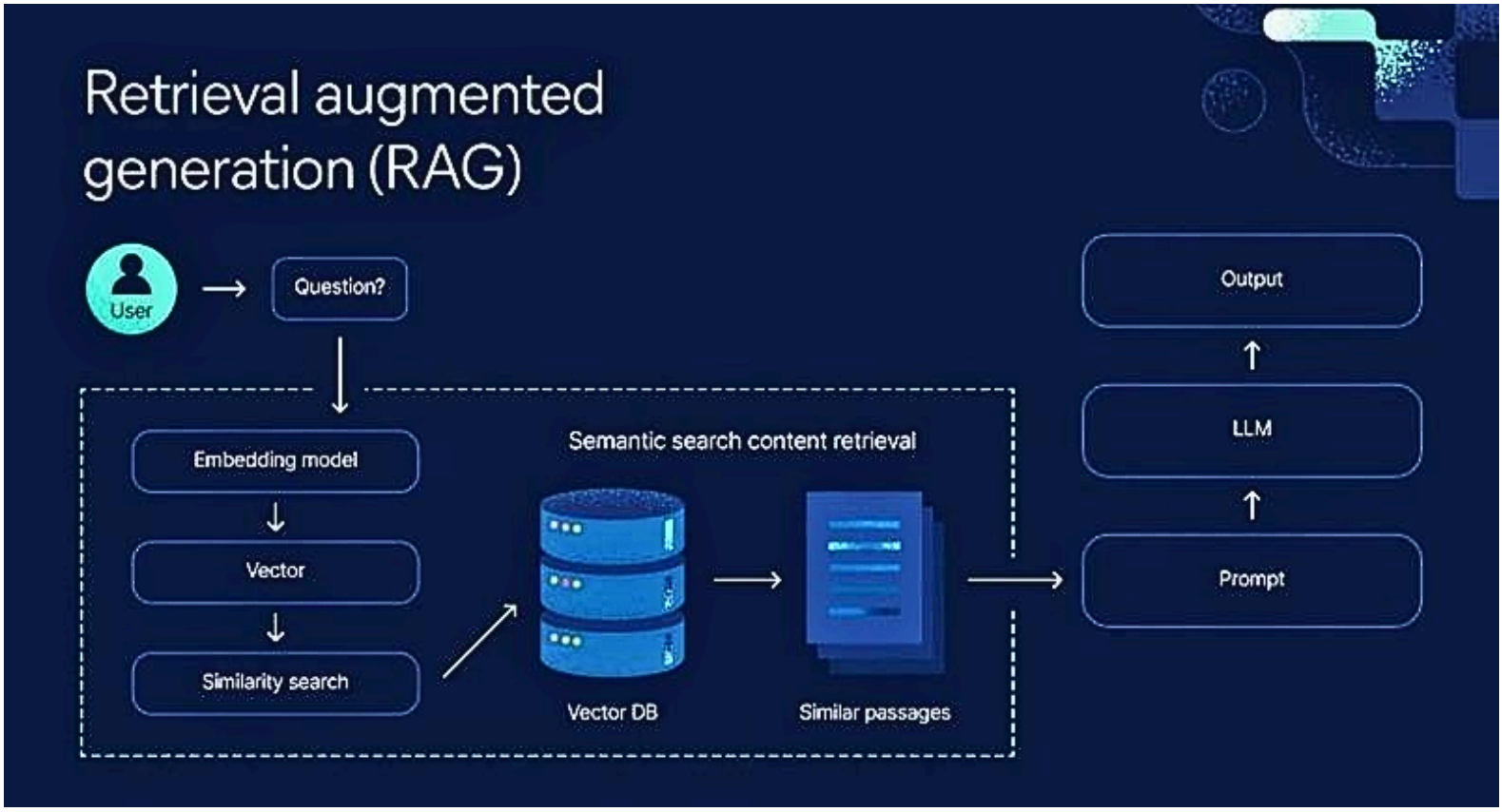


Figure 1: Retrieval-Augmented Generation (RAG) architecture used for semantic search and contextual SQL generation.

## System Overview

**The Tabbie-SQL Engine is designed as a hybrid AI system that integrates semantic understanding with intelligent query generation. The system processes natural language inputs, analyzes their semantic meaning, and determines the most efficient way to retrieve the requested data from the database.**

**Rather than relying solely on AI-generated queries, the system incorporates a semantic memory layer that stores previously solved query patterns. This allows Tabbie AI to reuse existing SQL logic when similar queries are detected, significantly reducing computation time and improving system efficiency.**

**Through this architecture, Tabbie AI provides a balance between speed, scalability, and AI-powered reasoning.**

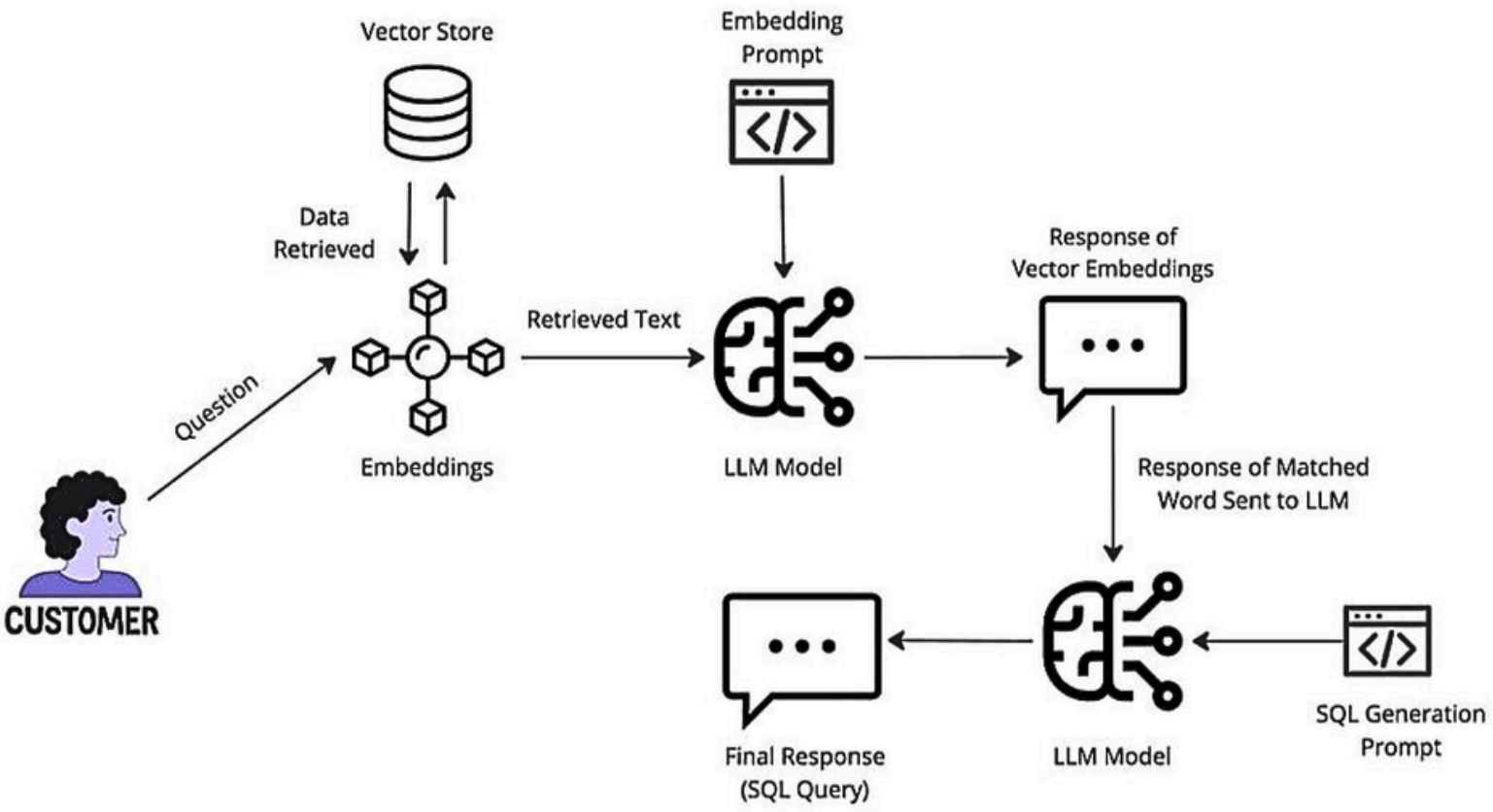


**Figure 2: Natural language query processing pipeline from user input to SQL generation.**

# Query Processing Workflow

**To ensure accurate and efficient query translation, the Tabbie-SQL Engine follows a structured multi-stage processing pipeline.**

**When a user submits a natural language query, the system first converts the input into a semantic vector representation. This allows the system to analyze the contextual meaning of the query rather than relying solely on keyword matching.**

**Next, the system performs a similarity search within the vector database to identify whether a similar query has been processed before. If a match is found, the stored SQL logic can be reused. Otherwise, the request is forwarded to the AI model to generate a new SQL query.**

**The following diagram illustrates the complete query processing workflow used in Tabbie AI**

## *System Workflow Overview*

User Query

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Semantic Embedding

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Vector Similarity Search (FAISS)

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Logic Guard Check

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Decision Engine

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Cached SQL LLM Generation

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Parameter Injection

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Database Execution

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Results to User

# Query Validation and Controlled Data Access

To ensure security, accuracy, and controlled access to sensitive data, the Tabbie-SQL Engine includes a validation layer that filters user queries before they are processed by the AI system.

## Domain Validation

The system is designed to answer only questions that are related to the organization's internal database. Before processing any query, the system analyzes the user's request to verify that it is relevant to the database schema and the available data.

If a user asks a question that is unrelated to the database — such as general knowledge questions or topics outside the system's scope — the query is rejected and the system returns a message indicating that the question is outside the supported domain.

This validation mechanism ensures that the AI assistant functions strictly as a database query assistant rather than a general-purpose chatbot.

## Controlled Table Access

Another important security feature is controlled table access. The system does not automatically allow the AI model to query every table in the database.

Instead, developers manually define a list of authorized tables that the system is allowed to access. The AI model can generate SQL queries only using these predefined tables.

By restricting the accessible tables, the system ensures that sensitive or irrelevant data sources remain protected and cannot be accessed by generated queries.

## Benefits of the Validation Layer.:

- *The validation mechanism provides several advantages:*
- *Prevents the system from answering unrelated or off-topic questions*
- *Ensures queries are limited to the organization's database domain*
- *Protects sensitive tables from unauthorized access*
- *Improves reliability and consistency of generated SQL queries*
- *Through this controlled validation process, Tabbie AI maintains both security and precision while interacting with enterprise databases.*

# Hybrid Semantic Memory Layer

**A core innovation of the Tabbie-SQL Engine is the implementation of a Hybrid Semantic Memory Layer.**

**This layer enables the system to detect logical similarities between user queries using vector embeddings. Instead of generating a new SQL query for every request, the system compares the semantic representation of the incoming query with previously stored query patterns.**

**If a match is detected, the engine can reuse the stored SQL logic and simply replace the relevant parameters. This approach significantly improves response time while maintaining the flexibility of AI-generated solutions when necessary.**

**By integrating semantic search with generative AI, Tabbie AI achieves a powerful balance between intelligent reasoning and computational efficiency.**

# *Chat System and Conversation History*

**In addition to its query processing capabilities, Tabbie AI provides a persistent chat interface that allows users to interact with the system through conversational sessions.**

**Each conversation is stored as part of a chat history, enabling users to review previous queries and results. This feature is particularly useful for analytical workflows where users may need to revisit earlier insights.**

**The system also supports multiple chat sessions, allowing users to create new chats while keeping previous conversations saved and accessible. This design ensures that users can organize their analytical tasks efficiently without losing previous interactions**

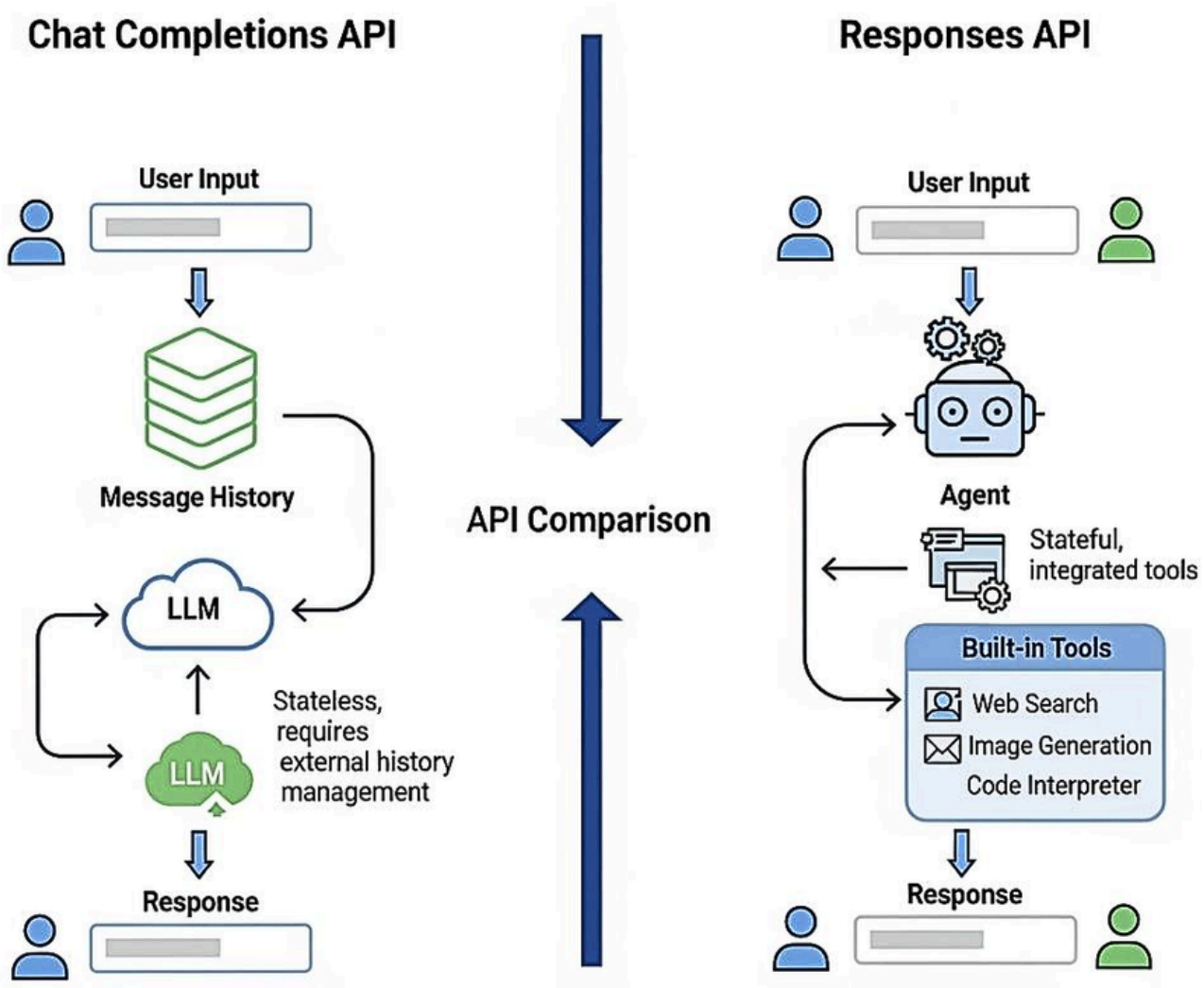


Figure 3: Comparison between chat-based interaction and agent-based response systems.

# Technical Stack

**The Tabbie-SQL Engine is built using a modern AI and data engineering stack designed to ensure high performance, scalability, and efficient query processing. Each component in the system plays a specific role in enabling natural language queries to be translated into accurate SQL statements.**

## 1- AI Reasoning Model – Google Gemini 2.5

**The system uses Google Gemini 2.5 as the primary large language model responsible for understanding natural language queries and generating the corresponding SQL statements. The model analyzes user input, interprets intent, and constructs syntactically correct SQL queries that can be executed on the database.**

## 2- Vector Database – FAISS

**To enable semantic similarity search, the system relies on FAISS (Facebook AI Similarity Search). FAISS is an industrial-grade vector database developed by Meta that allows efficient indexing and retrieval of high-dimensional vectors. It enables the system to quickly find previously solved queries that share similar semantic meaning.**

## 3- Embedding Model – HuggingFace all-MiniLM-L6-v2

**User queries are converted into vector representations using the HuggingFace all-MiniLM-L6-v2 embedding model. This model transforms text into high-dimensional semantic vectors that capture the contextual meaning of the query, allowing accurate similarity matching within the vector database.**

## 4- Database Connectivity – SQLAlchemy

**The system communicates with the database through SQLAlchemy, a powerful Python SQL toolkit and Object Relational Mapper (ORM). SQLAlchemy simplifies database interaction and ensures reliable execution of generated SQL queries.**

## 5- User Interface – Streamlit

**The user interface is built using Streamlit, which provides a lightweight and interactive environment for users to submit natural language queries and instantly view the generated results. Streamlit enables rapid development of data-driven web applications with minimal complexity.**

# *Future Improvements*

**Although Tabbie AI provides an effective solution for translating natural language queries into SQL statements, several enhancements can be implemented in future versions to further improve the system's capabilities.**

- ***Multi-Source Data Integration***

In the current version, Tabbie AI primarily retrieves information directly from databases using SQL queries. However, the system can be extended in future versions to support reading and analyzing data from multiple file formats such as:

PDF files

Excel spreadsheets

Word documents

CSV files

- ***User Account Management***

Another important future improvement for the system is the implementation of a User Account Management system. This would allow multiple users to access the system using their own individual login credentials.

With this feature, each user would be able to:

Log into the system using a personal username and password

Access only the data they are authorized to view

Save their query history and previous interactions

Continue their analytical work from earlier sessions

Additionally, system administrators would be able to manage user permissions and define which tables or datasets each user is allowed to access.

This feature would significantly improve security, access control, and organization, especially in enterprise environments where multiple users interact with the same database system

- ***Multi-Database Support***

Currently, the system uses Microsoft SQL Server as the primary database engine. Future versions could support additional database systems such as:

MySQL

PostgreSQL

Oracle

MongoDB

Supporting multiple database systems would increase the flexibility of the platform and allow it to operate in a wider range of enterprise environments.

- ***Data Visualization***

Another potential enhancement is the integration of data visualization tools that allow results to be displayed as:

Charts

Graphs

Interactive dashboards

Instead of presenting results only as tables, visual representations can help users better understand the data.

- ***Multi-Language Support***

Future versions of Tabbie AI could support multiple languages, allowing users to interact with the system using their preferred language.

This would make the platform more accessible and useful in international environments.